

The Top Open Source Cloud Tools

A cloud tool is a software program designed to make local and cloud based components work together. This article lists the top-of-the-line open source cloud tools in use today.

Enterprises are now using various IT resources to enhance their business. Profits increase when expenses decrease, and open source tools that are free to use and modify can help companies to reduce costs. Cloud vendors are also delivering numerous new tools to help enterprises get most value from cloud services. These tools enable organisations to use their budgets and manpower for more strategic business projects. Here is a list of the top open source tools in use today.

Grafana

Grafana allows you to query, visualise, and understand your metrics no matter where they are stored. You can create,

explore and share dashboards with your team, and foster a data driven culture. Its best use cases are infra time series data, industrial sensors, weather and process control. Grafana is written in Go and Node.js. It comes with an inbuilt embedded database called sqlite3. We can use other data stores like MySQL, PostgreSQL, Graphite, Influx DB, Prometheus and Elasticsearch for storing data. It also supports other data stores by using additional plugins.

Grafana can be installed on Linux, Windows, Docker and Mac. It comes with built-in user controls and authentication mechanisms like LDAP, Google Auth and GitHub, which help control access to the dashboards. Top of above listed authentications, Grafana

supports AD authentication as well. Grafana has an API interface that can be used to save a dashboard, create users and update data sources.

You can create dynamic and reusable dashboards with minimum effort and also have the option to create templates for re-use. Exploring and searching logs with preserved label filters makes your job easier for creating dashboards. You can even use ad-hoc queries to explore specific data and drill down to source. Grafana has an inbuilt alerting system with which you can define the alert rules for the specified metrics. It will continuously evaluate the same and send notifications to systems, in the form of an email, or via Slack, PagerDuty, VictorOps and OpsGenie.

When you compare Grafana with Kibana, the former is a better fit for time series data based on specific metrics (e.g., CPU/disk/IO utilisation). But Kibana is a better fit for analytics dashboards that use Elasticsearch capabilities. So we cannot use Grafana for data search and exploring.

Terraform

Terraform is a cloud immutable infrastructure tool from HashiCorp. It is used to build, change and version infrastructure efficiently. In short, Terraform is useful to deliver Infrastructure-as-a-Code (IAAC). It supports the infrastructure of most of the public cloud service providers as well as on-premise infrastructure.



Figure 1: Grafana dashboard